

Does Structured Triage Context Help a Chest-Radiograph Pneumonia Model? A Time-Safe, Multi-Seed, Externally-Validated Multimodal Study with Missing-Data Cautions

Yazan Al-Dabain · Arafat Md Easin · Mufti Mahmud · Swakkhar Shatabda

Received: date / Accepted: date

Abstract Background: Emergency-department (ED) triage records structured vital signs before chest imaging, motivating multimodal models that fuse the chest radiograph (CXR) with clinical context for pneumonia detection. Whether such fusion actually helps—under strict time-safety and rigorous evaluation—is unclear. **Methods:** We build a temporally safe cohort from MIMIC-CXR-JPG, MIMIC-IV and MIMIC-IV-ED, anchoring every feature at the CXR acquisition time t_0 and admitting only measurements resolved at or before t_0 . On a patient-level temporal split (test $n=1,075$), we train image-only, concatenation- and attention-fusion, and clinical-baseline models across five random seeds, adding an early-laboratory panel as a third modality. We report AUROC, AUPRC, expected calibration error (ECE) and Brier score as mean $\pm$ SD over seeds with paired per-seed deltas, external validation on NIH ChestX-ray14, and explicit leakage and missing-data audits. **Results:** A fine-tuned

DenseNet-121 reaches AUROC 0.737 ± 0.003 , far above triage-only baselines (0.61). Adding triage vitals leaves discrimination unchanged ($\Delta\text{AUROC} +0.004 \pm 0.009$, within a pre-specified ± 0.05 margin). A calibration advantage reported from a single checkpoint (ECE 0.040 vs 0.067) shrinks to $\Delta\text{ECE} -0.013 \pm 0.016$ across seeds—single-seed evaluation overstated it $\sim 2\times$. Adding early labs yields a small, seed-consistent AUROC gain ($+0.009 \pm 0.005$) that an ablation shows is reproduced exactly by lab *missingness indicators alone*, with no contribution from lab values, and is not attributable to leakage. Externally, discrimination transfers with minimal loss (AUROC 0.722 ± 0.005 on 112,120 NIH radiographs). **Conclusion:** The chest radiograph carries the discriminative signal; immediately available clinical context does not reliably add discrimination, and naive multimodal fusion can appear beneficial for reasons that will not generalise. We offer two transferable cautions for clinical multimodal AI: single-seed calibration inflation, and missingness masquerading as multimodal benefit.

Keywords Pneumonia detection · Chest radiograph · Multimodal fusion · Emergency department triage · Calibration · Reproducibility · Missing data · External validation

1 Introduction

Pneumonia is a leading cause of ED presentation and admission, and the frontal chest radiograph (CXR) remains central to its diagnosis [Metlay et al(2019)Metlay, Waterer, Long, et al.]. Deep-learning CXR classifiers approach radiologist-level performance on several findings, and a substantial literature proposes fusing imaging with structured clinical data [Acosta et al(2022)Acosta, Falcone, Rajpurkar, and Topol, et al.].

Yazan Al-Dabain
Department of Software Technology and Methodology, Faculty of Informatics, Eötvös Loránd University, Pázmány Péter Sétány 1/C, Budapest 1117, Hungary
E-mail:

Arafat Md Easin
Department of Software Technology and Methodology, Faculty of Informatics, Eötvös Loránd University, Pázmány Péter Sétány 1/C, Budapest 1117, Hungary
E-mail: arafatmd easin@inf.elte.hu

Mufti Mahmud
Department of Computer Science, Nottingham Trent Univer-
sity, NG11 8NS Nottingham, UK
E-mail: mufti.mahmud@ntu.ac.uk

Swakkhar Shatabda
Department of Computer Science and Engineering, BRAC
University, Dhaka 1212, Bangladesh
E-mail: swakkhar.shatabda@bracu.ac.bd

3 Results

3.1 The image dominates; fusion is discrimination-neutral

Across five seeds (Table 1, Fig. 1), both CXR models beat triage-only baselines by more than 12 AUROC points. Adding triage vitals does not change discrimination (concat – image $\Delta\text{AUROC} +0.004 \pm 0.009$; lower bound well inside the -0.05 non-inferiority margin). Attention fusion matches image AUROC but is severely and variably miscalibrated (ECE 0.076 ± 0.042 ; Fig. 2).

3.2 Caution 1: single-seed evaluation inflates the calibration effect

A single checkpoint suggested the concat model was markedly better calibrated (ECE 0.040 vs 0.067, a 40% reduction). Across seeds the image-only ECE alone ranges 0.041–0.060, and the paired advantage shrinks to $\Delta\text{ECE} -0.013 \pm 0.016$ (4/5 seeds favour concat; range crosses zero). The single-seed estimate overstated the benefit roughly twofold; the effect is also binning-dependent (Fig. 6) [Nixon et al(2019)Nixon, Dusenberry, Zeng, et al(2021)Zeng, et al(2021)]. Brier scores are essentially unchanged, so any effect is distributional alignment, not refinement.

3.3 Caution 2: the labs “benefit” is missing-data structure

Appending early labs gives the only seed-consistent AUROC gain (labs – image $+0.009 \pm 0.005$, 5/5 seeds, range excludes zero). Three analyses show this is neither chemistry nor leakage. *Ablation*: a model using lab *missingness flags only* (no values) reproduces the gain exactly (flags – image $+0.009$; Fig. 2); lab values add nothing further to discrimination. *Subcohort*: on the 269 test studies that actually have labs, discrimination is tied (labs 0.778 vs image 0.776). *Leakage audit*: 0 of 314,145 lab measurements post- t_0 ; 0 cross-split subjects; preprocessing fit on train only; the “any lab drawn” indicator has near-null univariate AUROC (0.464). The gain thus reflects the weak complementary signal of *whether* bloodwork was ordered, is small, and lies within the patient-level bootstrap CI.

3.4 Clinical availability of labs

Under time-safe matching, only 25.4% of ED-anchored CXR studies (272/1,075 test) have any resolved lab at or before imaging; CBC and metabolic panels

reach $\sim 25\%$, but CRP and procalcitonin—the most pneumonia-specific markers—are present in 0.5% and 0.6% (Fig. 4). At the moment of imaging, the informative labs are essentially unavailable.

3.5 Operating points, subgroups, modern backbone

At a fixed sensitivity of 0.90, specificity is 0.31 (image) vs 0.32 (concat); at 0.95, 0.17 vs 0.18—clinically equivalent. Subgroup AUROC reveals disparities not closed by fusion (Fig. 5): male 0.77 vs female 0.71; White 0.75, Black 0.71, Hispanic 0.64 (small strata); PA view 0.75 vs AP 0.72 [Seyyed-Kalantari et al(2021)Seyyed-Kalantari, Zhang, M, Larrazabal et al(2020)Larrazabal, Nieto, Peterson, Milone, and Ferraz, et al(2021)A CheXpert-trained DenseNet-121 [Cohen et al(2022)Cohen, Viviano, et al(2022)] used zero-shot (AUROC 0.667) or as a frozen-feature linear probe (0.687), does not outperform our task-tuned model (0.737).

3.6 External validation

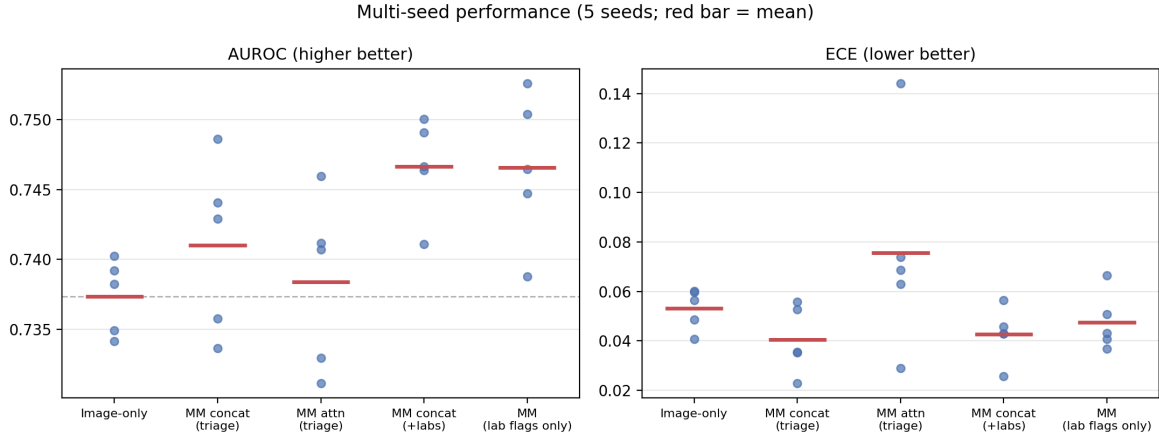
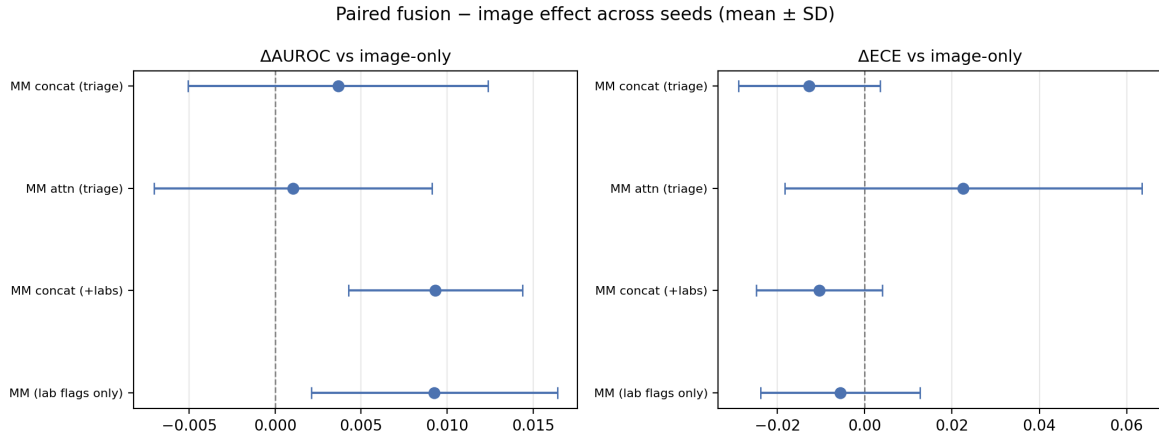
We evaluated the image models on all 112,120 radiographs of NIH ChestX-ray14 [Wang et al(2017)Wang, Peng, Lu, Lu, et al(2017)Zeng, et al(2021)] and chest11 [Wang et al(2017)Wang, Peng, Lu, Lu, et al(2017)] (a different institution, scanner population and label pipeline; 1,431 pneumonia-positive, prevalence 1.28%). Discrimination transfers with minimal loss: AUROC 0.722 ± 0.005 across five seeds (ensemble 0.726), only 0.016 below the internal 0.737 (Fig. 3). As expected under the large prevalence shift (45%→1.3%), prevalence-dependent metrics degrade (AUPRC 0.030, ECE 0.44) because probabilities are calibrated to the internal base rate; AUROC (prevalence-independent) is the appropriate cross-dataset metric, and it holds despite domain shift and NLP-derived external labels.

4 Discussion

The chest radiograph carries the discriminative signal for radiographic pneumonia in this ED cohort; immediately available structured clinical context—triage vitals, and even a broad early-lab panel—does not add reliable discrimination, and the image model generalises externally with minimal loss. This is a well-powered negative answer to a specific, clinically motivated question, accompanied by two transferable methodological cautions. Reporting calibration or small discrimination effects from a single training run can overstate them (here $\sim 2\times$); multi-seed replication should be standard. And where clinical inputs are heavily and non-randomly missing, a multimodal model can exploit missingness structure and present a spurious benefit that neither

Table 1 Multi-seed test-set performance (mean $\pm$ SD over 5 seeds; $n=1,075$; prevalence 45.3%). Clinical baselines are single-fit.

Model	AUROC	AUPRC	ECE	Brier
Image-only DenseNet-121	0.737 ± 0.003	0.719 ± 0.004	0.053 ± 0.008	0.207 ± 0.001
Multimodal — concat (triage)	0.741 ± 0.006	0.715 ± 0.005	0.040 ± 0.014	0.205 ± 0.002
Multimodal — attention (triage)	0.738 ± 0.006	0.713 ± 0.012	0.076 ± 0.042	0.212 ± 0.010
Multimodal — concat (+labs)	0.747 ± 0.004	0.724 ± 0.004	0.043 ± 0.011	0.203 ± 0.002
Logistic regression (triage)	0.606	0.548	0.037	0.242
XGBoost (triage)	0.611	0.567	0.046	0.241

**Fig. 1** Per-model AUROC and ECE across five seeds (points, seeds; red bar, mean). Discrimination is stable; the image-only ECE alone varies substantially by seed.**Fig. 2** Paired fusion — image effects across seeds (mean $\pm$ SD). Every effect straddles zero; the labs and lab-flags-only deltas are identical, indicating the labs effect is driven by missingness rather than lab values.

reflects the measured values nor generalises—a failure mode distinct from data leakage, which our audits explicitly excluded.

Limitations. Training and multimodal evaluation are single-institution (MIMIC) and retrospective; external validation confirms the image branch generalises to NIH but is necessarily limited to imaging, as no external dataset links ED triage to the CXR, and probabilities require recalibration to the target base rate before deployment (the `u_ignore` policy also raises internal

evaluation prevalence to 45.3%). The lab-present sub-cohort is small ($n=269$); the modern-backbone comparison uses frozen features rather than end-to-end fine-tuning; and NIH labels are NLP-derived, adding label noise to the external estimate.

5 Conclusion

Under strict time-safety and multi-seed evaluation, fusing immediately available clinical context with the chest

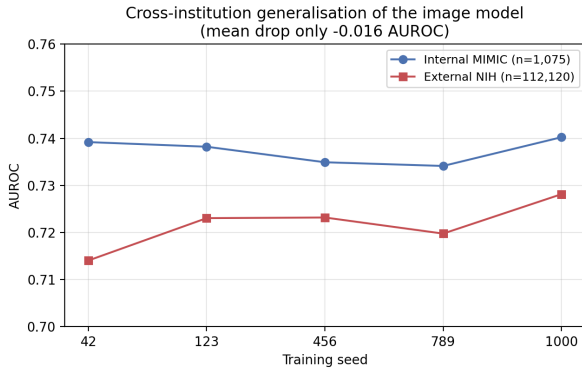


Fig. 3 Cross-institution generalisation: per-seed AUROC on internal MIMIC ($n=1,075$) vs external NIH ChestX-ray14 ($n=112,120$); mean drop -0.016 .

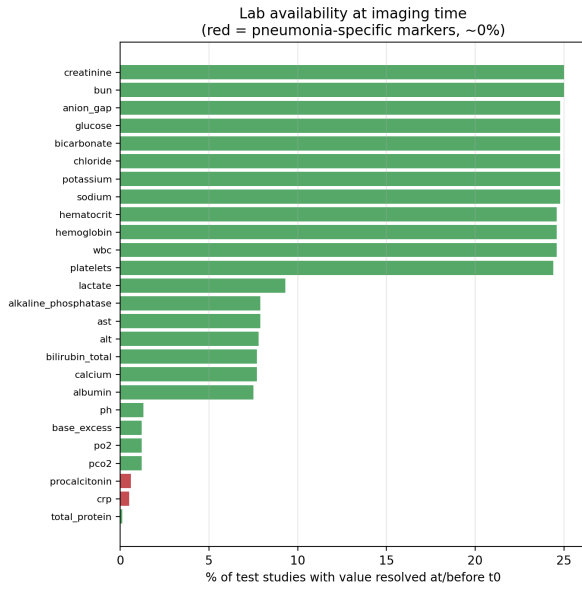


Fig. 4 Fraction of test studies with each lab resolved at or before t_0 ; pneumonia-specific markers (red) are essentially unavailable at imaging time.

radiograph does not reliably improve pneumonia discrimination, and the one apparent gain is an artefact of missing-data structure. The image model itself is robust and generalises across institutions. Beyond the negative result, the study offers two reusable cautions—single-seed calibration inflation and missingness-as-signal—for the growing field of clinical multimodal AI.

Declarations

Acknowledgements. The authors thank the MIT Laboratory for Computational Physiology and PhysioNet for maintaining and distributing the MIMIC databases, and the U.S. National Institutes of Health for the ChestX-ray14 dataset.

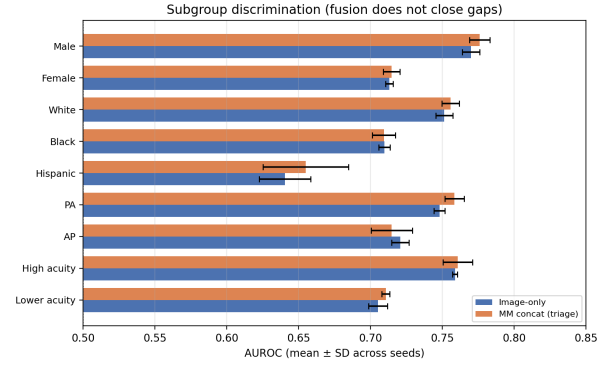


Fig. 5 Subgroup AUROC (mean $\pm$ SD across seeds) for image-only vs concat fusion; fusion does not close sex, race or view gaps.

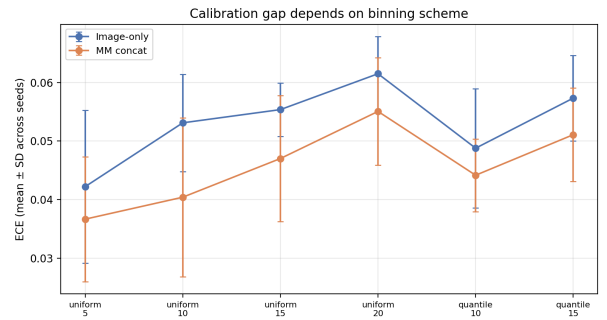


Fig. 6 ECE under different binning schemes; the image-concat calibration gap depends on the binning choice.

Authors' contributions. YAD designed and implemented the data pipeline, trained all models, and performed the analyses. AME conceptualised and supervised the study and contributed to the manuscript. MM and SS contributed to methodology, interpretation, and critical revision. All authors read and approved the final manuscript.

Funding. The authors received no specific funding for this work.

Conflict of interest. The authors declare no competing interests.

Ethics approval. All data are publicly available and de-identified (MIMIC under PhysioNet's credentialed Data Use Agreement; NIH ChestX-ray14 public release). No additional ethical approval was required; procedures followed the Declaration of Helsinki.

Data and code availability. MIMIC and NIH ChestX-ray14 are publicly available (MIMIC via PhysioNet credentialing). All pipeline, training, and evaluation code is available at <https://github.com/wa5rx3/MultiM>

odalPneumonia. Every reported number is regenerated by a named script tied to the fixed temporal split.

References

- Acosta et al(2022)Acosta, Falcone, Rajpurkar, and Topol. Acosta JN, Falcone GJ, Rajpurkar P, Topol EJ (2022) Multimodal biomedical AI. *Nature Medicine* 28(9):1773–1784, DOI 10.1038/s41591-022-01981-2
- Chen and Guestrin(2016). Chen T, Guestrin C (2016) XG-Boost: A scalable tree boosting system. In: *KDD*, pp 785–794, DOI 10.1145/2939672.2939785
- Cohen et al(2022)Cohen, Viviano, Bertin, Morrison, Torabian, Cohen JP, Viviano JD, Bertin P, Morrison P, Torabian P, Guarrera M, Lungren MP, Chaudhari A, Brooks R, Hashir M, Bertrand H (2022) TorchXRyVision: A library of chest X-ray datasets and models. In: *Proceedings of Machine Learning Research (MIDL)*, vol 172, pp 231–249
- Collins et al(2024)Collins, Moons, Dhiman et al. Collins GS, Moons KGM, Dhiman P, et al (2024) TRI-POD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 385:e078,378, DOI 10.1136/bmj-2023-078378
- Guo et al(2017)Guo, Pleiss, Sun, and Weinberger. Guo C, Pleiss G, Sun Y, Weinberger KQ (2017) On calibration of modern neural networks. In: *International Conference on Machine Learning (ICML)*, vol 70, pp 1321–1330
- Hayat et al(2022)Hayat, Geras, and Shamout. Hayat N, Geras KJ, Shamout FE (2022) MedFuse: Multi-modal fusion with clinical time-series data and chest X-ray images. In: *Machine Learning for Healthcare (MLHC)*, PMLR, vol 182, pp 479–503
- Huang et al(2017)Huang, Liu, van der Maaten, and Weinberger. Huang G, Liu Z, van der Maaten L, Weinberger KQ (2017) Densely connected convolutional networks. In: *CVPR*, pp 4700–4708, DOI 10.1109/CVPR.2017.243
- Irvin et al(2019)Irvin, Rajpurkar, Ko et al. Irvin J, Rajpurkar P, Ko M, et al (2019) CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In: *AAAI*, vol 33, pp 590–597, DOI 10.1609/aaai.v33i01.3301590
- Johnson et al(2019)Johnson, Pollard, Berkowitz, Greenbaum, Lungren, Deng, Mark, and Horng. Johnson AEW, Pollard TJ, Berkowitz SJ, Greenbaum NR, Lungren MP, Deng Cy, Mark RG, Horng S (2019) MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs. *arXiv:1901.07042*, URL <https://arxiv.org/abs/1901.07042>
- Johnson et al(2023a)Johnson, Bulgarelli, Pollard, Horng, Celi, and Mark. Johnson AEW, Bulgarelli L, Pollard TJ, Horng S, Celi LA, Mark RG (2023a) MIMIC-IV-ED: Emergency department module of MIMIC-IV. URL <https://physionet.org/content/mimic-iv-ed/>
- Johnson et al(2023b)Johnson, Bulgarelli, Shen et al. Johnson AEW, Bulgarelli L, Shen L, et al (2023b) MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data* 10(1):1, DOI 10.1038/s41597-022-01899-x
- Larrazabal et al(2020)Larrazabal, Nieto, Peterson, Milone, and Ferrante. Larrazabal AJ, Nieto N, Peterson V, Milone DH, Ferrante E (2020) Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis. *PNAS* 117(23):12,592–12,594, DOI 10.1073/pnas.1919012117
- Metlay et al(2019)Metlay, Waterer, Long et al. Metlay JP, Waterer GW, Long AC, et al (2019) Diagnosis and treatment of adults with community-acquired pneumonia (ats/idsa guideline). *American Journal of Respiratory and Critical Care Medicine* 200(7):e45–e67, DOI 10.1164/rccm.201908-1581ST
- Nixon et al(2019)Nixon, Dusenberry, Zhang, Jerfel, and Tran. Nixon J, Dusenberry MW, Zhang L, Jerfel G, Tran D (2019) Measuring calibration in deep learning. In: *CVPR Workshops*, URL <https://arxiv.org/abs/1904.01685>
- Piaggio et al(2012)Piaggio, Elbourne, Pocock, Evans, and Altman. Piaggio G, Elbourne DR, Pocock SJ, Evans SJW, Altman DG (2012) Reporting of noninferiority and equivalence randomized trials: Extension of the CONSORT 2010 Statement. *JAMA* 308(24):2594–2604, DOI 10.1001/jama.2012.87802
- Seyyed-Kalantari et al(2021)Seyyed-Kalantari, Zhang, McDermott, Chen, and Seyyed-Kalantari L, Zhang H, McDermott MBA, Chen IY, Ghassemi M (2021) Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine* 27:2176–2182, DOI 10.1038/s41591-021-01595-0
- Soenksen et al(2022)Soenksen, Ma, Zeng, Boussioux, Villalobos Carballo, N Soenksen LR, Ma Y, Zeng C, Boussioux L, Villalobos Carballo K, Na L, Wiberg HM, Li ML, Fuentes I, Bertsimas D (2022) Integrated multimodal artificial intelligence framework for healthcare applications. *npj Digital Medicine* 5:149, DOI 10.1038/s41746-022-00689-4
- Tejani et al(2024)Tejani, Klontzas, Gatti, Mongan, Moy, Park, and Kahn. Tejani AS, Klontzas ME, Gatti AA, Mongan JT, Moy L, Park SH, Kahn CE (2024) Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence* 6(4):e240,300, DOI 10.1148/ryai.240300
- Wang et al(2017)Wang, Peng, Lu, Lu, Bagheri, and Summers. Wang X, Peng Y, Lu L, Lu Z, Bagheri M, Summers RM (2017) ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp 2097–2106, DOI 10.1109/CVPR.2017.369